

Pharmacy Module API Endpoints

MediX Backend System

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HTTP Status Codes

All endpoints return appropriate HTTP status codes:

- **200 OK** for successful operations
- **201 Created** for successful POST (though Spring returns 200 by default)
- **404 Not Found** for non-existent resources
- **400 Bad Request** for validation errors
- **500 Internal Server Error** for server issues

1 Medicine Endpoints

1.1 Create Medicine

- **Method:** POST
- **URL:** `http://localhost:8080/api/medicines`
- **Request Body:**

```
1 {  
2     "company": "Pfizer",  
3     "medicineName": "Lipitor",  
4     "genericName": "Atorvastatin",  
5     "quantity": 100,  
6     "unitCost": 5.50,  
7     "unitPrice": 12.99,  
8     "expiryDate": "2025-12-31"  
9 }
```

1.2 Get All Medicines

- Method: GET
- URL: `http://localhost:8080/api/medicines`
- Response:

```
1 [
2   {
3     "id": 1,
4     "company": "Pfizer",
5     "medicineName": "Lipitor",
6     "genericName": "Atorvastatin",
7     "quantity": 100,
8     "unitCost": 5.5,
9     "unitPrice": 12.99,
10    "expiryDate": "2025-12-31"
11  }
12 ]
```

1.3 Get Medicine by ID

- Method: GET
- URL: `http://localhost:8080/api/medicines/1`
- Response:

```
1 {
2   "id": 1,
3   "company": "Pfizer",
4   "medicineName": "Lipitor",
5   "genericName": "Atorvastatin",
6   "quantity": 100,
7   "unitCost": 5.5,
8   "unitPrice": 12.99,
9   "expiryDate": "2025-12-31"
10 }
```

1.4 Get Medicine by Name

- Method: GET
- URL: `http://localhost:8080/api/medicines/name/Lipitor`
- Response:

```
1 {
2   "id": 1,
3   "company": "Pfizer",
4   "medicineName": "Lipitor",
5   "genericName": "Atorvastatin",
```

```

6     "quantity": 100,
7     "unitCost": 5.5,
8     "unitPrice": 12.99,
9     "expiryDate": "2025-12-31"
10 }

```

1.5 Update Medicine

- Method: PUT
- URL: `http://localhost:8080/api/medicines/1`
- Request Body:

```

1 {
2     "company": "Pfizer Updated",
3     "medicineName": "Lipitor Updated",
4     "genericName": "Atorvastatin Calcium",
5     "quantity": 150,
6     "unitCost": 6.00,
7     "unitPrice": 13.50,
8     "expiryDate": "2026-01-31"
9 }

```

1.6 Delete Medicine

- Method: DELETE
- URL: `http://localhost:8080/api/medicines/1`

1.7 Get Medicines by Company

- Method: GET
- URL: `http://localhost:8080/api/medicines/company/Pfizer`
- Response:

```

1 [
2     {
3         "id": 1,
4         "company": "Pfizer",
5         "medicineName": "Lipitor",
6         "genericName": "Atorvastatin",
7         "quantity": 100,
8         "unitCost": 5.5,
9         "unitPrice": 12.99,
10        "expiryDate": "2025-12-31"
11    }
12 ]

```

1.8 Get Medicines by Generic Name

- Method: GET
- URL: `http://localhost:8080/api/medicines/generic/Atorvastatin`
- Response:

```
1 [
2   {
3     "id": 1,
4     "company": "Pfizer",
5     "medicineName": "Lipitor",
6     "genericName": "Atorvastatin",
7     "quantity": 100,
8     "unitCost": 5.5,
9     "unitPrice": 12.99,
10    "expiryDate": "2025-12-31"
11  }
12 ]
```

1.9 Get Expired Medicines

- Method: GET
- URL: `http://localhost:8080/api/medicines/expired`
- Response:

```
1 [
2   {
3     "id": 1,
4     "company": "Pfizer",
5     "medicineName": "Lipitor",
6     "genericName": "Atorvastatin",
7     "quantity": 100,
8     "unitCost": 5.5,
9     "unitPrice": 12.99,
10    "expiryDate": "2025-12-31"
11  }
12 ]
```

2 Bill Endpoints

2.1 Create Bill

- Method: POST
- URL: `/api/bills`
- Request Body:

```

1 {
2   "customerName": "John Doe",
3   "phoneNumber": "123-456-7890",
4   "date": "2023-10-15",
5   "prescriptionId": null,
6   "subTotal": 45.00,
7   "tax": 4.50,
8   "total": 49.50,
9   "sellType": false,
10  "billItems": [
11    {
12      "medicineName": "Aspirin",
13      "quantity": 10,
14      "unitPrice": 1.00,
15      "discount": 0.50,
16      "total": 9.50
17    },
18    {
19      "medicineName": "Paracetamol",
20      "quantity": 5,
21      "unitPrice": 2.00,
22      "discount": 1.00,
23      "total": 9.00
24    }
25  ]
26 }

```

2.2 Get All Bills

- Method: GET
- URL: /api/bills

2.3 Get Bill by ID

- Method: GET
- URL: /api/bills/{id}

2.4 Get Bills by Date

- Method: GET
- URL: /api/bills/date/{date}

2.5 Get Bills by Customer Name

- Method: GET
- URL: /api/bills/customer/{customerName}

2.6 Get Bills by Phone Number

- Method: GET
- URL: /api/bills/phone/{phoneNumber}

2.7 Get Bill Items by Bill ID

- Method: GET
- URL: /api/bills/{id}/items

2.8 Delete Bill

- Method: DELETE
- URL: /api/bills/{id}

2.9 Bill History

- Method: GET
- URL: http://localhost:8080/api/bills/history
- Response:

```
1  [
2    {
3      "billId": 1,
4      "date": "2024-01-15",
5      "prescriptionId": 101,
6      "patientId": 5001,
7      "patientPhone": "+1234567890",
8      "sellType": false,
9      "total": 49.50
10   },
11   {
12     "billId": 2,
13     "date": "2024-01-14",
14     "prescriptionId": null,
15     "patientId": null,
16     "patientPhone": "+1987654321",
17     "sellType": true,
18     "total": 125.75
19   },
20   {
21     "billId": 3,
22     "date": "2024-01-13",
23     "prescriptionId": 102,
24     "patientId": 5002,
25     "patientPhone": "+1555666777",
26     "sellType": false,
```

```
27     "total": 87.25
28   }
29 ]
```

2.10 Revenue Analytics

- Method: GET
- URL: `http://localhost:8080/api/bills/revenue/analytics`
- Response:

```
1 {
2   "todayRevenue": 1250.75,
3   "weeklyRevenue": 8750.50,
4   "monthlyRevenue": 32500.25
5 }
```

3 Pharmacist Endpoints

3.1 Create Pharmacist

- Method: POST
- URL: `/api/pharmacists`
- Request Body:

```
1 {
2   "name": "John Pharmacist",
3   "email": "john.pharmacist@medix.com",
4   "phoneNumber": "+1234567890",
5   "password": "pharmacist123",
6   "address": "123 Pharmacy Street, Medical City"
7 }
```

- Response:

```
1 {
2   "id": 2503001,
3   "name": "John Pharmacist",
4   "email": "john.pharmacist@medix.com",
5   "phoneNumber": "+1234567890",
6   "password": "pharmacist123",
7   "address": "123 Pharmacy Street, Medical City",
8   "doctor": null
9 }
```

3.2 Get Pharmacist by ID

- Method: GET
- URL: /api/pharmacists/2503001
- Response:

```
1 {  
2   "id": 2503001,  
3   "name": "John Pharmacist",  
4   "email": "john.pharmacist@medix.com",  
5   "phoneNumber": "+1234567890",  
6   "password": "pharmacist123",  
7   "address": "123 Pharmacy Street, Medical City",  
8   "doctor": null  
9 }
```

- Error Response:

```
1 {  
2   "error": "Pharmacist not found",  
3   "pharmacistId": 9999999  
4 }
```

3.3 Login Pharmacist

- Method: POST
- URL: /api/pharmacists/by-email
- Request Body:

```
1 {  
2   "email": "john.pharmacist@medix.com",  
3   "password": "pharmacist123"  
4 }
```

- Success Response:

```
1 {  
2   "success": true,  
3   "message": "Pharmacist found",  
4   "data": {  
5     "id": 2503001,  
6     "name": "John Pharmacist",  
7     "email": "john.pharmacist@medix.com",  
8     "phoneNumber": "+1234567890",  
9     "password": "pharmacist123",  
10    "address": "123 Pharmacy Street, Medical City",  
11    "doctor": null  
12  }  
13 }
```


3.4 Update Pharmacist

- Method: PUT
- URL: /api/pharmacists/2503001
- Request Body:

```
1 {  
2   "name": "John Updated Pharmacist",  
3   "phoneNumber": "+0987654321",  
4   "address": "456 New Pharmacy Ave, Medical City"  
5 }
```

- Response:

```
1 {  
2   "id": 2503001,  
3   "name": "John Updated Pharmacist",  
4   "email": "john.pharmacist@medix.com",  
5   "phoneNumber": "+0987654321",  
6   "password": "pharmacist123",  
7   "address": "456 New Pharmacy Ave, Medical City",  
8   "doctor": null  
9 }
```

3.5 Delete Pharmacist

- Method: DELETE
- URL: /api/pharmacists/2503001
- Success Response:

```
1 {  
2   "success": true,  
3   "message": "Pharmacist deleted successfully"  
4 }
```